

Abdullah Ahmad Khan

Applied Artificial Intelligence | Data Science | Trustworthy AI

Perth, Western Australia

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Academic Profile

Early-career academic and final-stage PhD researcher in Computer Science specialising in **artificial intelligence, machine learning, data science, trustworthy AI, and multimodal learning**. Experienced in applied AI research, interdisciplinary collaboration, reproducible experimentation, and undergraduate/postgraduate teaching. Two first-author journal papers are accepted in *Neural Networks* and *Neurocomputing*; thesis submission is expected in September 2026.

Education

Doctor of Philosophy in Computer Science

Murdoch University

Thesis submission expected Sep 2026

Perth, Western Australia

- Research focus: trustworthy and privacy-preserving AI, machine unlearning, multimodal and vision-language systems, and robust model evaluation.
- Supervisors: Assoc. Prof. Mohammed Kaosar, Prof. Hamid Laga, Prof. Ferdous Sohel.

Master of Computer Science and Applications (MCA)

Aligarh Muslim University

2021

India

- Relevant study covered AI, machine learning, algorithms, databases, software engineering, networks, cybersecurity, and numerical/statistical computing.

Academic Appointments and Teaching

Academic Lecturer

University of Notre Dame Australia

2026–Present

Perth, Western Australia

- Deliver lectures, tutorials, and practical sessions in **COMP6012 Cyber Threat Intelligence** and **COMP6013 Network Security and Cryptography**.
- Develop learning activities and assessments, mark student work, provide consultations, and support applied laboratory and case-based learning.

Casual Academic, Information Technology

Murdoch University

2024–Present

Perth, Western Australia

- Deliver **420+ teaching hours per year** across **ICT619 Artificial Intelligence**, **ICT206 Intelligent Systems**, and **ICT100 Transition to IT**, supporting **200+ undergraduate and postgraduate students**.
- Lead tutorials, laboratories, workshops, programming activities, demonstrations, and guided problem solving; mark assessments and contribute to moderation and academic-integrity processes.
- Mentor students on project proposals, early research drafts, Python/ML experiments, data preprocessing, model evaluation, and technical writing.

Research Experience

PhD Researcher: Applied Data Science & Trustworthy AI

Murdoch University, School of Information Technology

2023–Present

Perth, Western Australia

- Conduct independent and collaborative research in machine learning, multimodal AI, machine unlearning, privacy-preserving ML, model robustness, and vision-language model evaluation.
- Develop reproducible AI pipelines for data preparation, model training, evaluation, attack simulation, robustness testing, and ablation studies using Python, PyTorch, TensorFlow, scikit-learn, and Hugging Face.
- Investigate privacy leakage, membership inference, quantization-induced recovery, metric unreliability, and post-deployment model behavior using statistical testing, bootstrap analysis, repeated-seed experiments, and model comparison.
- Lead first-author projects from research question and method design through implementation, experimentation, analysis, manuscript preparation, reviewer response, and research-artifact release.
- Collaborate across computer science and applied genomics, translating machine-learning methods to biomedical prediction and privacy-regulation problems.

Applied and Cross-Disciplinary Research

- **Antimicrobial resistance prediction:** applied machine-unlearning methods under GDPR right-to-erasure requirements, linking AI with biomedical prediction, privacy, and governance.
- **Quantisation-induced recovery:** studied whether low-bit quantisation can restore information after machine unlearning, providing a deployment-aware view of trustworthy AI.
- **Multimodal and vision-language AI:** evaluate image-text models using behavioural, privacy, representation-level, and robustness measures.

Selected Publications and Research Outputs

- **Khan, A. A.**, et al. (2026). *Selective Multimodal Unlearning via Boundary-Based Forgetting*. *Neural Networks*. **Accepted**.
- **Khan, A. A.**, et al. (2026). *Hessian-Guided Gradient Unlearning*. *Neurocomputing*. **Accepted**.
- **Khan, A. A.**, & Sohel, F. (2026). *DurableUn: Quantisation-Induced Recovery Attacks in Machine Unlearning*. arXiv:2605.02196.
- **Khan, A. A.**, et al. (2026). *Metric Unreliability in Multimodal Machine Unlearning*. arXiv:2605.02206.
- Saniya, & **Khan, A. A.** (2026). *Machine Unlearning for GDPR Right-to-Erasure in Antimicrobial Resistance Prediction Models*. DOI: 10.64898/2026.03.09.26347960.
- **Khan, A. A.**, Belal, M., & Ullah, G. (2023). *Rapid Face Mask Detection and Person Identification Model Based on Deep Neural Networks*. Springer. DOI: 10.1007/978-981-19-8136-4_10.
- Belal, M., Ullah, G., & **Khan, A. A.** (2022). *Islamophobic Tweet Detection using Transfer Learning*. *IEEE CSI*. DOI: 10.1109/CSI54720.2022.9923957.

Student Supervision and Research Mentoring

- Supervise and mentor undergraduate and postgraduate students on project proposals, early research drafts, research questions, literature reviews, method selection, experimental design, and evaluation planning.
- Provide technical guidance in Python, machine learning, data preprocessing, model evaluation, reproducibility, and technical writing; prepared to contribute to Honours and HDR co-supervision.

Industry and Software Development Experience

Freelance Programmer / Software Developer

2021–Present

Freelance

Remote

- Deliver software implementation, debugging, feature development, technical documentation, and client-focused technical solutions.

Technical and Academic Skills

AI and data science: machine learning, deep learning, multimodal learning, vision-language models, NLP, computer vision, trustworthy AI, privacy-preserving ML.

Programming and computing: Python, R, SQL, Java, C++, NumPy, pandas, scikit-learn, PyTorch, TensorFlow, Hugging Face, Git/GitHub, Linux, GPU experimentation.

Research methods: experimental design, statistical testing, bootstrap analysis, ablation studies, robustness evaluation, reproducible research, manuscript writing, peer-review revision.

Teaching: lectures, tutorials, laboratories, blended learning, assessment design, marking, moderation, consultation, and learning-resource development.

Leadership, Service and Engagement

- **Founder & President, Murdoch Cricket Club** (2024–Present): lead a 50+ member student organisation, including budgeting, sponsorship, events, and stakeholder engagement.
- **President, Murdoch University Muslim Students Association** (2023–Present): lead community support, university liaison, advocacy, and student engagement.

Awards, Work Rights

Murdoch International Postgraduate Scholarship; University Merit Scholarship; Central Government Scholarship; State Government Scholarship.

Work rights: Full Australian working rights.

Referees

Available upon request.